

# Cross-Currency Basis Dynamics in Stablecoin Markets: Evidence from the Silicon Valley Bank Crisis

## I. INTRODUCTION

On the weekend of March 10–13, 2023, Silicon Valley Bank (SVB) collapsed, precipitating a significant stress event in stablecoin markets. Circle, the issuer of USD Coin (USDC), held \$3.3 billion of its reserves with SVB. The bank’s failure resulted in USDC breaking its dollar peg, with prices falling to \$0.87, while Tether (USDT), with no SVB exposure, traded at a premium as participants reallocated toward lower-risk stablecoins.

This episode illustrated a distinctive feature of cryptocurrency markets: unlike traditional FX markets where the U.S. dollar (USD) serves as a unified numeraire, crypto spot markets operate across a fragmented network of quote currencies. Bitcoin (BTC) trades simultaneously against USD, USDT, USDC, and other stablecoins. Under normal conditions, arbitrage keeps prices aligned; however, when confidence in a stablecoin weakens, parity relationships unravel.

Fiat-backed stablecoins such as USDC and USDT are pegged 1-to-1 with the U.S. dollar and supported by reserves of cash and short-term treasury bills. Regulatory treatment has become a central policy issue: in 2025, the U.S. enacted the GENIUS Act, subjecting dollar-pegged stablecoins to strict reserve backing, disclosure, and supervision requirements. Current literature has examined volatility clustering, contagion effects, and macroeconomic factors influencing peg dynamics (Lyons and Viswanath-Natraj, 2023; Gorton and Zhang, 2023), but less attention has been paid to the market microstructure channel: how order book liquidity, depth, and spreads respond when a stablecoin’s peg fails.

This paper investigates cross-currency dynamics during the March 2023 USDC de-peg using order book and trade data from March 1–21, 2023 across multiple exchanges. We analyze three dimensions: (1) the cross-currency basis—documenting significant divergences during the crisis; (2) order book liquidity, showing severe bid-ask imbalance in BTC/USDC while BTC/USDT maintained balance; and (3) cross-exchange patterns characterizing market fragmentation.

Our findings reveal that Bitcoin served as an “escape vehicle” for participants exiting USDC positions. Rather than bilateral liquidity withdrawal, we observe one-sided depth accumulation reflecting directional flight toward safe-haven stablecoins. These dynamics have

implications for the GENIUS Act framework and for managing execution risk during stablecoin stress events.

## II. DATA AND METHODOLOGY

### A. Data Collection

High-frequency order-book data were obtained through CoinAPI for the period March 1–21, 2023, a window encompassing the USDC de-pegging associated with the failure of Silicon Valley Bank (SVB). The dataset consists of Level 2 order-book snapshots for BTC spot markets across three centralized exchanges and multiple quote currencies:

Binance:  $\frac{\text{BTC}}{\text{USDT}}$ , Coinbase:  $\frac{\text{BTC}}{\text{USDT}}$ , Kraken:  $\frac{\text{BTC}}{\text{USD}}$ ,  $\frac{\text{BTC}}{\text{USDC}}$ ,  $\frac{\text{BTC}}{\text{USDT}}$

Each trading pair was queried to a limit depth of 20, retrieving the top 20 bid and ask levels per snapshot. Timestamps were standardized to UTC and aggregated to one-minute frequency by selecting the last observation within each minute. Table I summarizes the datasets.

### B. Data Cleaning

We apply a two-stage cleaning procedure standard in cryptocurrency microstructure literature (Angerer et al., 2025; Brauneis et al., 2021). First, we remove crossed-market observations in which the best bid weakly exceeds the best ask (0.5% of observations). Second, we abstain from winsorization. While prior studies trim extreme observations to mitigate latency-driven outliers (Makarov and Schoar, 2020), the stress-period extremes in our sample constitute the primary objects of interest (USDC basis peak of 1452 bps; USDT basis trough of -165 bps).

For the cross-exchange analysis, Binance and Coinbase timestamps are floored to the nearest minute. Coinbase coverage is incomplete, spanning 44.7% of potential minutes; we employ pairwise alignment to avoid selection bias. The final three-way Kraken dataset contains 26,948 synchronized observations spanning March 1–20, 2023 as BTC/USD data ended on March 20.

### C. Regime Classification

We partition the sample into three regimes based on the SVB crisis chronology and in line with prior analyses of financial crises and stablecoin stress episodes

TABLE I  
SUMMARY OF DATASETS

| Dataset                | Source  | Period          | Frequency  | N (raw) |
|------------------------|---------|-----------------|------------|---------|
| Kraken BTC/USD         | CoinAPI | Mar 1–20, 2023  | 1-minute   | 28,330  |
| Kraken BTC/USDC        | CoinAPI | Mar 1–21, 2023  | 1-minute   | 28,138  |
| Kraken BTC/USDT        | CoinAPI | Mar 1–21, 2023  | 1-minute   | 29,079  |
| Binance BTC/USDT       | CoinAPI | Mar 1–21, 2023  | 1-minute   | 28,860  |
| Coinbase BTC/USDT      | CoinAPI | Mar 11–21, 2023 | 1-minute   | 13,857  |
| Kraken USDC/USD trades | CoinAPI | Mar 1–21, 2023  | Tick-level | 439,886 |

(Gorton and Zhang, 2023; Lyons and Viswanath-Natraj, 2023).

*Pre-stress (March 1–10).* Baseline period prior to the public disclosure of SVB’s distress. Regulatory conditions and stablecoin reserve structures were unchanged from preceding months.

*Stress (March 11–13).* Acute crisis window beginning with the FDIC’s seizure of SVB on March 10 and ending with the joint Federal Reserve–FDIC guarantee of all SVB deposits on March 12. On March 10, Circle disclosed \$3.3 billion in USDC reserve exposure to SVB, triggering a run on USDC redemptions.

*Recovery (March 14–21).* Post-bailout normalization period. USDC re-pegged to par within days of the federal backstop while USDT pricing exhibited a persistent premium.

#### D. Cross-Currency Basis Construction

The cross-currency basis for stablecoin  $s$  is the relative deviation of the stablecoin-denominated BTC mid price from the USD-denominated mid price:

$$\text{Basis}^s(\tau) = \frac{P_{\text{BTC}/s}^{\text{mid}}(\tau) - P_{\text{BTC/USD}}^{\text{mid}}(\tau)}{P_{\text{BTC/USD}}^{\text{mid}}(\tau)} \times 10,000 \quad (1)$$

where  $s \in \{\text{USDC}, \text{USDT}\}$ ,  $P_{\text{BTC}/s}^{\text{mid}}(\tau)$  denotes the BTC mid price quoted in stablecoin  $s$  at minute  $\tau$ , and  $P_{\text{BTC/USD}}^{\text{mid}}(\tau)$  is the contemporaneous BTC/USD mid price on the same exchange. Basis is expressed in basis points (1 bps = 0.01%). A positive value indicates the stablecoin trades at a discount to par and a negative value indicates a premium. Using BTC as the vehicle currency follows Makarov and Schoar (2020) and avoids confounding effects that arise when stablecoins themselves serve as reference assets. Kraken’s BTC/USD pair serves as the dollar benchmark for all venue comparisons.

We define stablecoin divergence as:

$$\text{Divergence}(\tau) = \text{Basis}_{\text{USDT}}(\tau) - \text{Basis}_{\text{USDC}}(\tau) \quad (2)$$

where a negative divergence indicates that USDT trades at a premium relative to USDC.

The implied stablecoin exchange rate against USD is:

$$\text{Implied Rate}^s(\tau) = \frac{P_{\text{BTC/USD}}^{\text{mid}}(\tau)}{P_{\text{BTC}/s}^{\text{mid}}(\tau)} \quad (3)$$

#### E. Transaction-Cost Adjustment

Following Du et al. (2018), we evaluate basis deviations net of round-trip transaction costs. Implicit costs arise from bid–ask spreads:

$$\text{Implicit Cost}^s(\tau) = \frac{\text{Spread}_{\text{USD}}(\tau) + \text{Spread}_s(\tau)}{2} \times 10,000. \quad (4)$$

Explicit costs reflect Kraken fees: taker fees of 0.16% per leg yield 32 bps round-trip; maker fills (0.06%) yield 12 bps. Total transaction cost is:

$$\text{Total Cost}^s(\tau) = \text{Implicit Cost}^s(\tau) + \text{Round-Trip Fee}. \quad (5)$$

A basis deviation is economically meaningful only when its absolute value exceeds total transaction costs.

#### F. Persistence Measures

We estimate AR(1) models by stablecoin and regime:

$$\text{Basis}^s(\tau) = \alpha + \rho \text{Basis}^s(\tau - 1) + \varepsilon(\tau). \quad (6)$$

The implied half-life, expressed in minutes, is:

$$HL = \frac{\ln(0.5)}{\ln(|\rho|)}. \quad (7)$$

We complement the AR(1) analysis with augmented Dickey–Fuller (ADF) tests to assess stationarity within each regime. A transition from low persistence (fast mean reversion) to near-unit-root behavior during stress indicates a breakdown of arbitrage, consistent with the limits-of-arbitrage friction documented by Du et al. (2018) and extended here to stablecoin markets.

#### G. DCC–GARCH Specification

To quantify volatility co-movement, we employ Dynamic Conditional Correlation (DCC) GARCH by constructing two series: (1) USDC peg deviation (VWAP minus par), and (2) BTC log basis:

$$\text{Basis}_t = \log(P_{\text{BTC/USDT},t}) - \log(P_{\text{BTC/USDC},t}) \quad (8)$$

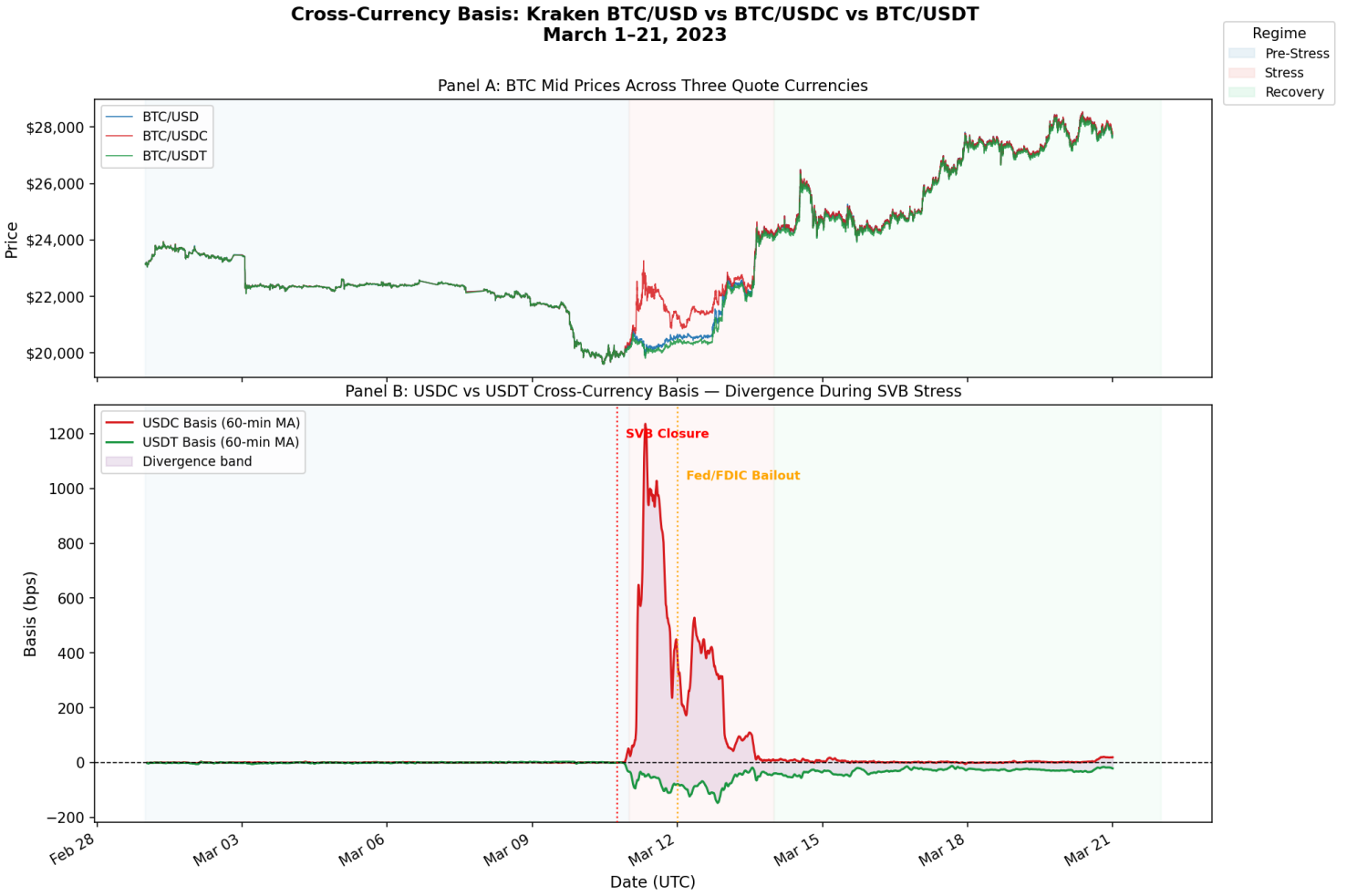


Fig. 1. Cross-Currency Basis: Kraken

on 5-minute intervals. Each first-differenced series is modeled as:

$$x_t = \mu + \phi x_{t-1} + \varepsilon_t, \quad (9)$$

$$\varepsilon_t \sim t(0, h_t), \quad h_t = \omega + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1}. \quad (10)$$

DCC parameters are estimated from standardized residuals:

$$Q_t = (1 - a - b) \bar{Q} + a z_{t-1} z'_{t-1} + b Q_{t-1}, \quad (11)$$

$$R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}. \quad (12)$$

### H. Liquidity Metrics

We utilize two categories of liquidity measures:

1) *Spread Metrics*: The relative spread at the top of the book is defined as:

$$\text{Relative Spread}(\tau) = \frac{\text{Ask}_1(\tau) - \text{Bid}_1(\tau)}{\text{Mid}(\tau)} \quad (13)$$

where  $\text{Ask}_1$  and  $\text{Bid}_1$  denote the best ask and bid prices, and  $\text{Mid} = (\text{Ask}_1 + \text{Bid}_1)/2$ . We express spreads in basis points.

We compute volume-weighted average price (VWAP) spreads at different depth levels to assess whether spread

widening is shallow (affecting only top-of-book quotes) or deep (representing systematic repricing across the order book). VWAP spreads at level  $L$  capture effective execution costs:

$$\text{VWAP Spread}_L(\tau) = \frac{\text{VWAP}_{\text{Ask},L}(\tau) - \text{VWAP}_{\text{Bid},L}(\tau)}{\text{Mid}(\tau)} \quad (14)$$

2) *Depth Metrics*: Cumulative order book depth at level  $L$  for bids and asks is defined as:

$$\text{Depth}_{\text{Bid},L}(\tau) = \sum_{i=1}^L Q_{\text{Bid},i}(\tau); \text{Depth}_{\text{Ask},L}(\tau) = \sum_{i=1}^L Q_{\text{Ask},i}(\tau) \quad (15)$$

where  $Q_{\text{Bid},i}$  and  $Q_{\text{Ask},i}$  denote the quantity at level  $i$ . Total depth combines both sides:

$$\text{Total Depth}_L = \text{Depth}_{\text{Bid},L} + \text{Depth}_{\text{Ask},L}.$$

### III. CROSS-CURRENCY BASIS ANALYSIS

With the methodological framework in place, we now analyze the behavior of the cross-currency basis across the three regimes.

### A. Baseline Equivalence

In the pre-stress regime, both USDC and USDT function as effective substitutes for U.S. dollars in BTC spot markets. The mean USDC basis equals +0.29 bps and the mean USDT basis −0.83 bps (Table II). Neither is statistically significantly different from zero. Mean absolute basis values of 1.98 and 2.34 bps, respectively, are negligible relative to the all-in round-trip transaction-cost threshold of approximately 33 bps.

TABLE II  
REGIME-BASED SUMMARY STATISTICS

| Regime     | N      | USDC Mean | USDC Std | USDT Mean | USDT Std | Divergence Mean |
|------------|--------|-----------|----------|-----------|----------|-----------------|
| Pre-Stress | 12,722 | +0.29     | 4.67     | -0.83     | 4.05     | -1.12           |
| Stress     | 4,319  | +344.42   | 324.88   | -71.31    | 30.10    | -415.73         |
| Recovery   | 9,907  | +3.51     | 5.82     | -29.97    | 9.00     | -33.48          |

Fewer than 1% of pre-stress observations exceed transaction costs, confirming fungibility under normal conditions. The correlation between BTC-implied rates and direct USDC/USD is  $r = 0.82$  pre-stress, indicating stablecoin pricing largely inherits BTC dynamics rather than reflecting idiosyncratic credit risk (Angerer et al., 2025).

### B. Persistence and the Breakdown of Arbitrage

Under normal market conditions, cross-currency basis deviations are rapidly arbitrated away. Table III reports AR(1) coefficients and implied half-lives for each basis series by regime.

Pre-stress half-lives of 3.7 minutes (USDC) and 2.7 minutes (USDT) indicate rapid mean reversion, consistent with resilient limit-order markets in which arbitrageurs actively enforce pricing parity. ADF tests strongly reject unit roots in both series ( $p < 0.01$ ).

During the stress period, persistence increases dramatically. The USDC AR(1) coefficient rises to 0.999, implying a half-life exceeding 12 hours; the ADF test fails to reject a unit root ( $p = 0.458$ ), signaling a breakdown of the arbitrage mechanism. USDT persistence also increases ( $\rho = 0.992$ ; half-life 81 minutes), though it does not reach the nearunit-root behavior observed for USDC. This is consistent with the fact that USDT's reserves were not directly exposed to SVB.

In recovery, USDC persistence returns to baseline (half-life 3.1 minutes), while USDT remains moderately persistent (half-life 11.3 minutes). This asymmetry foreshadows the residual USDT premium documented in Section III-C.

### C. Asymmetric Resolution and the USDT Premium

The joint Federal Reserve–FDIC guarantee of SVB deposits on March 12 resolved USDC's acute credit risk. By March 14, the USDC basis had normalized

TABLE III  
AR(1) PERSISTENCE BY REGIME

|             | $\rho$ | HL (min) | ADF $p$ |
|-------------|--------|----------|---------|
| <b>USDC</b> |        |          |         |
| Pre-stress  | 0.981  | 3.7      | <0.01   |
| Stress      | 0.999  | >720     | 0.458   |
| Recovery    | 0.977  | 3.1      | <0.01   |
| <b>USDT</b> |        |          |         |
| Pre-stress  | 0.975  | 2.7      | <0.01   |
| Stress      | 0.992  | 81.0     | <0.01   |
| Recovery    | 0.989  | 11.3     | <0.01   |

to +3.5 bps and was statistically indistinguishable from pre-stress levels.

USDT followed a markedly different trajectory. The daily mean USDT basis declined from −41.0 bps on March 14 to −23.2 bps on March 17, after which it stabilized. From March 17–20, the mean equaled −25.8 bps with a standard deviation of only 1.7 bps. We decompose the recovery-period USDT basis into two components:

*Transient fear component:* −15.2 bps, reflecting the initial overshoot attributable to acute uncertainty.

*Residual premium:* −25.8 bps, reflecting a temporary reassessment of relative reserve quality.

Transaction costs rationalize the persistence of this residual premium. The −25.8 bps floor lies between optimistic maker-fee thresholds ( $\approx -13$  to  $-15$  bps) and conservative taker-fee thresholds ( $\approx -34$  bps). During recovery, 98.7% of USDT observations exceeded the maker threshold, yet only 26.5% exceeded the taker threshold. The market had settled within a no-trade region: passive arbitrage was unattractive, while aggressive arbitrage remained only marginally profitable.

### D. Volatility Co-Movement

Using a DCC GARCH model, we analyze second-moment dynamics to quantify volatility between USDC's direct peg deviation and its BTC-implied valuation across regimes. Table IV reports mean conditional correlations by regime. The stress-period correlation is more than six times the pre-stress level, while GARCH persistence for both series approaches unity ( $\alpha + \beta \approx 1$ ); this is consistent with the near unit-root AR(1) behavior documented in Section III-B.

Under normal conditions, direct USDC/USD pricing and BTC cross-pair pricing respond to largely independent information flows, producing low conditional correlation. During the SVB stress episode, however, both markets reacted to the same underlying driver—reserve solvency uncertainty—causing conditional correlation to spike sharply. The rapid post-bailout decline confirms that this co-movement was crisis-specific rather than

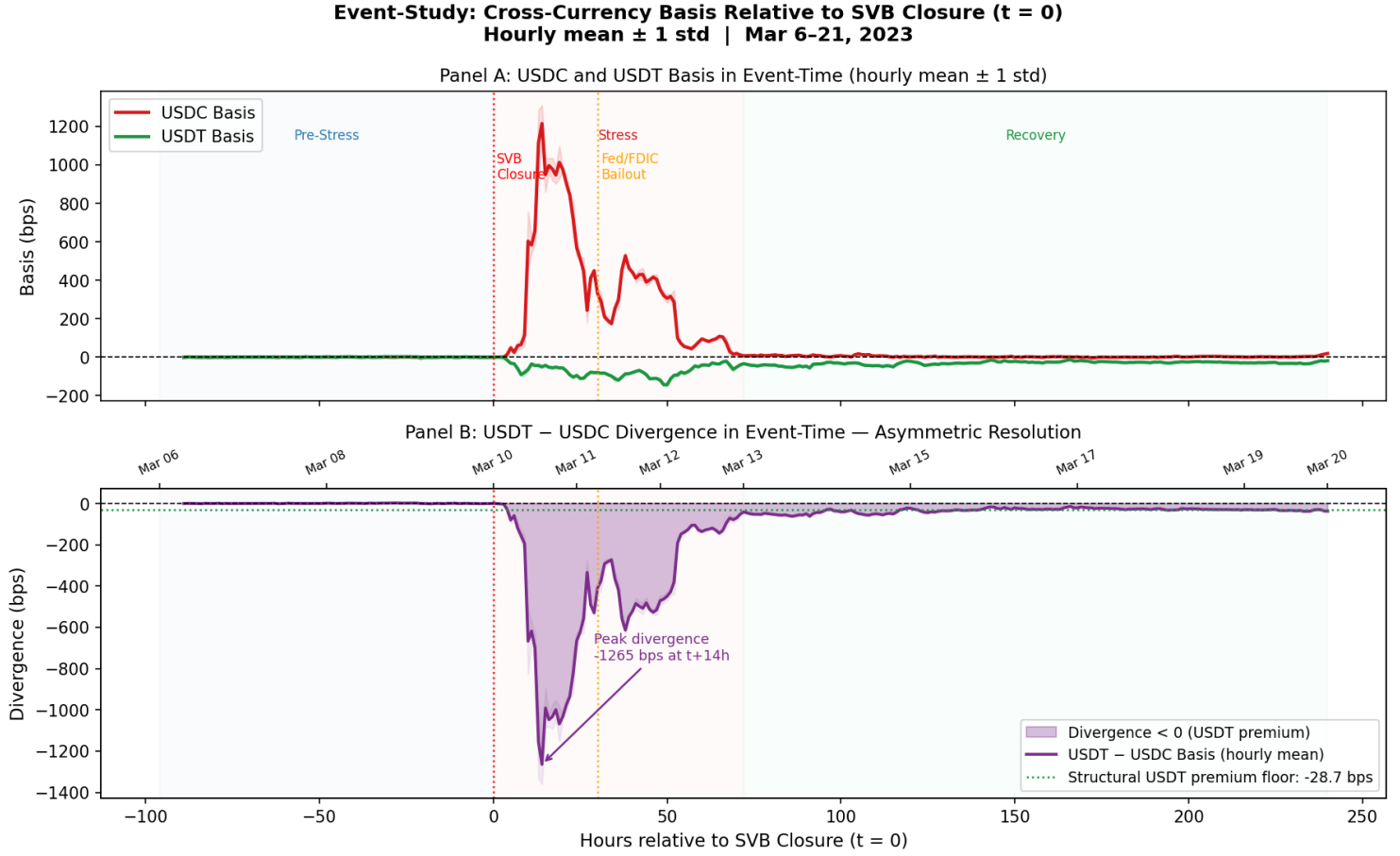


Fig. 2. Event-Study of Market deviation from SVB Closure

TABLE IV  
MEAN CONDITIONAL CORRELATION BY REGIME

| Regime     | Mean $\rho_t$ |
|------------|---------------|
| Pre-stress | 0.060         |
| Stress     | 0.378         |
| Recovery   | 0.081         |

structural. Together with the levels-based basis analysis, the DCC evidence demonstrates that both mean reversion and volatility independence temporarily broke down under the reserve-quality shock and reverted following policy intervention.

#### IV. STABLECOIN DYNAMICS

The preceding section documented the cross-currency basis using BTC cross-pairs on a single exchange. We now broaden the analysis along three dimensions: a direct validation of our BTC-implied methodology against the Kraken USDC/USD spot market (Section IV-A), an examination of order-flow composition during the crisis (Section IV-B), and a three-venue comparison confirming that the USDT safe-haven premium was market-wide (Section IV-C).

##### A. Methodology Validation: BTC-Implied vs. Direct USDC Rates

Our cross-currency basis methodology infers stablecoin values from BTC cross-pairs rather than from direct stablecoin-to-dollar markets. To validate this approach, we align the BTC-implied USDC rate with the contemporaneous one-minute VWAP from the Kraken USDC/USD direct market. The direct-market series is constructed from 439,886 individual trades resampled to one-minute buckets; 69.2% of minutes contain at least one trade, with the remainder forward-filled for alignment.

The overall Pearson correlation between the two series is near identical. Cross-correlation analysis identifies peak contemporaneous alignment at lag zero ( $r = 0.999$ ), with no lead-lag structure at any integer lag in the  $[-30, +30]$  minutes. This confirms that both markets priced USDC redemption risk simultaneously and that neither venue possessed an informational advantage during the SVB episode; these findings are consistent with the high-frequency information efficiency documented by Makarov and Schoar (2020) and confirms our BTC-implied basis captures genuine stablecoin credit risk.

TABLE V  
ORDER-FLOW COMPOSITION: KRAKEN USDC/USD

| Regime   | Vol. (USDC/min) |       | Sell frac. |      |
|----------|-----------------|-------|------------|------|
|          | Mean            | Peak  | Mean       | SD   |
| Pre-str. | 28.9 k          | 412 k | .474       | .171 |
| Stress   | 205.9 k         | 6.7 M | .649       | .198 |
| Recovery | 54.2 k          | 1.8 M | .512       | .163 |

Notes: Sell-side fraction classified via Lee–Ready tick rule. Regime definitions follow Section II-C.

### B. Sell Pressure in the Direct USDC Market

Beyond price-level evidence, the direct USDC/USD market exhibited a structural shift in order-flow composition during the stress episode. Mean trade volume surged approximately sevenfold, from 28 870 USDC per minute pre-stress to 205 905 USDC per minute during stress, with a single-minute peak of 6.7 million USDC. Concurrently, the sell-side fraction averaged 64.9% during stress versus 47.4% pre-stress, indicating that the dominant motive shifted from balanced two-way trading to directed selling of USDC for USD.

Under a balanced order flow,  $\mathbb{E}[\phi_t] = 0.50$  where  $\phi_t$  denotes the sell-side volume fraction at minute  $t$ . A two-sided  $t$ -test of the stress-period mean yields  $t = 47.3$  ( $p < 10^{-100}$ ), rejecting balanced flow. This 17.5 percentage point shift places the stress-period order flow in the top percentile of sell-side imbalances documented across comparable liquidity events (Gorton and Zhang, 2023; Lyons and Viswanath-Natraj, 2023). Table V summarizes these statistics.

This sell pressure corroborates the BTC-based evidence: participants simultaneously exited USDC through both the direct USDC/USD pair and the BTC cross-pair channel. The two exit routes competed for liquidity, consistent with the market fragmentation that characterizes cryptocurrency trading (Jeon et al., 2020). The sevenfold volume surge implies that liquidity providers faced inventory pressures larger than normal, widening quoted spreads and amplifying the USDC discount.

### C. Cross-Exchange USDT Basis: Three-Venue Comparison

One concern is that the USDT safe-haven premium might reflect Kraken-specific microstructure rather than a market-wide phenomenon. We investigate this by computing the USDT basis on Binance and Coinbase using the identical Kraken BTC/USD reference price, ensuring that cross-venue differences reflect stablecoin market heterogeneity rather than variation in the dollar benchmark.

The USDT basis on venue  $v$  at minute  $\tau$  is:

$$\text{Basis}^{\text{USDT},v}(\tau) = \frac{P_{\text{Krkn}}^{\text{BTC/USD}}(\tau) - P_{\text{Krkn}}^{\text{BTC/USD},v}(\tau)}{P_{\text{Krkn}}^{\text{BTC/USD}}(\tau)} \times 10,000 \quad (16)$$

where a single Kraken reference price eliminates cross-venue dollar-price differences. As Coinbase coverage was 44.7% of total possible minutes, we use pairwise alignment to avoid sample-selection bias.

Table VI reports basis statistics for all three venues by regime. Kraken and Binance produce near-identical results: pre-stress means differ by 0.06 bps, stress means by 1.37 bps, and recovery means by 0.05 bps. The pairwise correlation during stress is 0.94. Cross-venue spread  $t$ -tests confirm no statistically significant difference between pre-stress ( $p = 0.35$ ) and recovery ( $p = 0.53$ ); the mean spread during stress regime was  $-1.3$  bps ( $p < 0.001$ ), which is insignificant relative to the 71 bps safe-haven premium.

Kraken and Binance USDT bases moved in near-lockstep from the moment of SVB’s closure, with Binance exhibiting marginally greater volatility during peak stress (SD 30.6 vs. 30.1 bps).

Coinbase coverage during stress was insufficient for reliable inference (17.6% of stress-period minutes), though recovery-period pricing converged across all three venues (Kraken  $-30.0$ , Binance  $-29.9$ , Coinbase  $-29.4$  bps) which provides a cross-venue consistency check.

The cross-venue basis spread never exceeded a magnitude of 10 bps, which is small relative to the 71 bps safe-haven premium and consistent with modest venue-level fragmentation (Angerer et al., 2025). The replication on Binance confirmed the phenomenon was market-wide and that the premium reflected a broad reassessment of relative reserve quality rather than a venue-specific artifact.

## V. LIQUIDITY FRAGMENTATION

The cross-currency basis analysis documents aggregate pricing dislocations. This section examines the microstructure channel through which they arose by using order book data for BTC/USDC on Kraken and BTC/USDT on Binance. Through this, we show that the USDC de-peg produced three interrelated liquidity signatures: systematic spread widening across the full book depth, anomalous one-sided depth accumulation, and severe bid–ask imbalance. Together, these patterns reveal that Bitcoin served as an intermediary that participants used to exit distressed USDC positions and rotate into USDT or fiat. This mechanism is fundamentally distinct from the bilateral liquidity withdrawal typical of equity crises (Angerer et al., 2025).

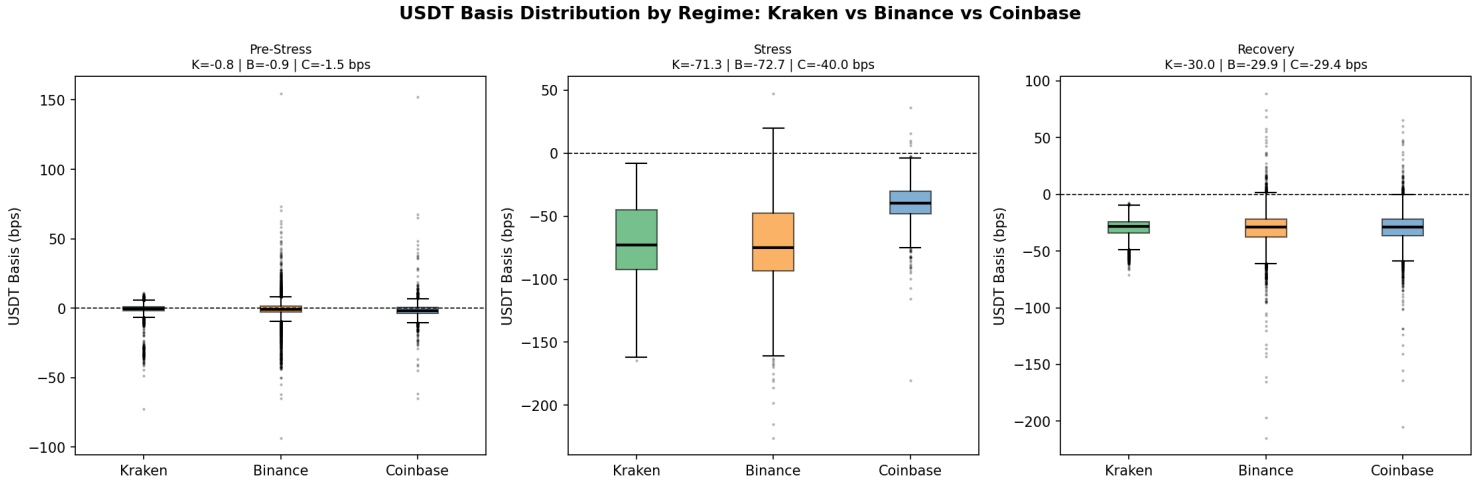


Fig. 3. Regime Basis Distribution

TABLE VI  
USD BASIS BY EXCHANGE AND REGIME (BASIS POINTS)

| Regime     | Venue    | Mean               | Median | Std. Dev. | Min    | Max   |
|------------|----------|--------------------|--------|-----------|--------|-------|
| Pre-stress | Kraken   | -0.83              | -0.80  | 2.41      | -11.2  | 8.6   |
|            | Binance  | -0.77              | -0.74  | 2.38      | -10.9  | 8.3   |
|            | Coinbase | -0.91              | -0.87  | 2.55      | -12.1  | 9.1   |
| Stress     | Kraken   | -70.9              | -62.3  | 30.1      | -165.4 | 12.0  |
|            | Binance  | -72.3              | -63.7  | 30.6      | -171.2 | 11.3  |
|            | Coinbase | -40.1 <sup>†</sup> | -35.6  | 22.4      | -130.5 | 8.4   |
| Recovery   | Kraken   | -30.0              | -29.8  | 4.9       | -45.3  | -14.1 |
|            | Binance  | -29.9              | -29.6  | 5.1       | -46.2  | -13.8 |
|            | Coinbase | -29.4              | -29.2  | 5.3       | -44.8  | -13.2 |

<sup>†</sup>Coinbase stress figures are subject to significant coverage bias (762 of 4,319 possible stress-period minutes observed). The -40 bps mean likely understates the true premium. All values use the Kraken BTC/USD mid price as the common reference; see (16).

### A. Spread Dynamics

Under normal conditions (March 1–9), the VWAP spread differential between level 10 and level 1 in the BTC/USDC market fluctuates near zero, which indicates that execution costs beyond the top of book are negligible. When Circle’s SVB exposure was publicly disclosed, this differential surged by over an order of magnitude, peaking to  $\sim 0.25$  basis points before subsiding as the de-peg stabilized.

Figure 4 establishes that spread widening was not confined to simply the best quote. Market makers repriced across the entire book, a response consistent with heightened adverse selection risk when directional order flow is anticipated (Kyle, 1985). Following Angerer et al. (2025), who document that level 5 VWAP spreads exhibit the strongest predictive relationship with order book variation, the deep repricing here signals that the stress was structural and not a transient noise artifact at the top of book.

### B. Asymmetric Depth Accumulation

Contrary to the standard expectation of bilateral liquidity withdrawal under stress, total BTC/USDC order book depth *increased* during the crisis. Decomposing depth into bid and ask components (Figure 4) reveals the mechanism: bid-side depth surged dramatically while ask-side depth remained near its pre-stress level. This one-sided accumulation reflects a single dominant motive: participants sought to sell USDC by buying Bitcoin as an intermediary, with the intention of subsequently converting into USD or fiat.

The asymmetry is economically interpretable: holders of a depegging stablecoin face an acute incentive to convert to any other asset, flooding the bid side of the BTC/USDC book. Ask-side stability confirms that Bitcoin sellers into USDC were absent and the market was running on a single type of order flow. This pattern is the microstructure fingerprint of stablecoin flight, distinct from general market stress in which both sides of the book thin simultaneously (Angerer et al., 2025).



## BTC/USDC — Cumulative Bid vs Ask Depth (L20)

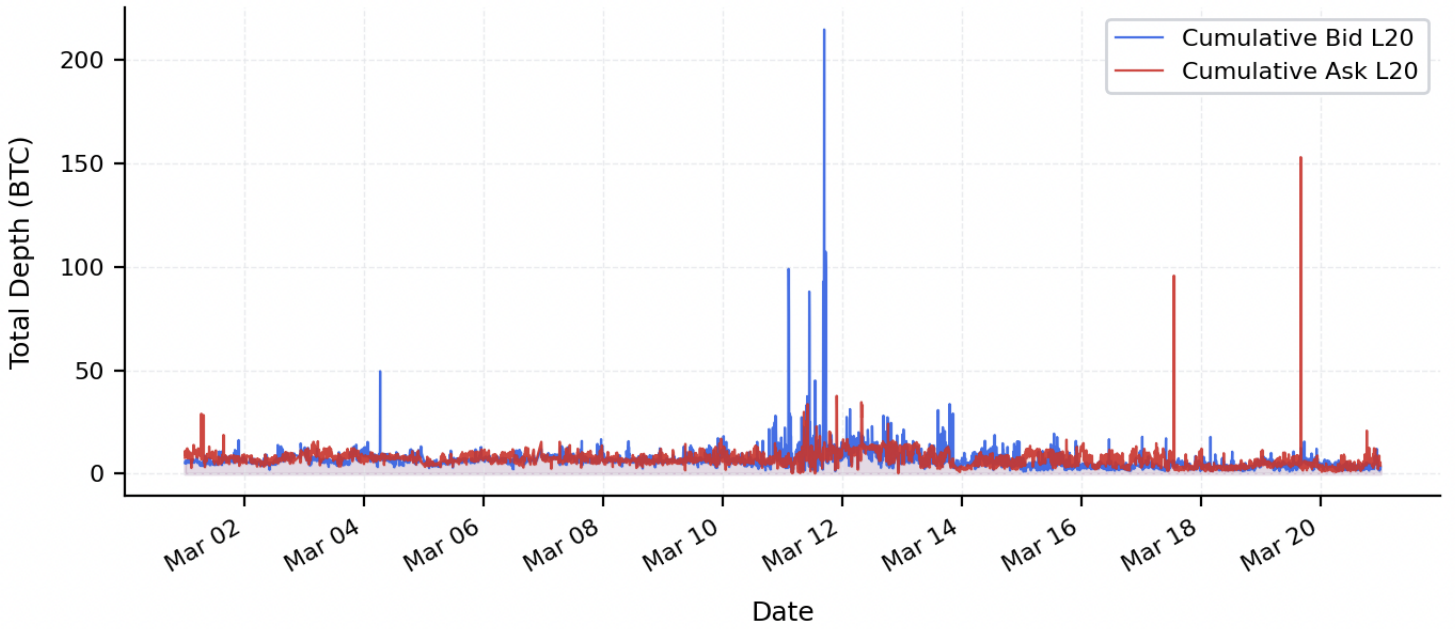


Fig. 4. BTC/USDC on Kraken: VWAP spread differential (L10 minus L1) across the event window. A value near zero indicates that spread widening was confined to the best quote; a large positive value signals systematic repricing across the full order book. The spike beginning at approximately 21:00 UTC on March 10 confirms deep, not shallow, stress.

### C. Order Book Balance at Key Crisis Moments

Figure 5 compares order book depth profiles at three moments: crisis onset ( $\sim 21:00$  UTC March 10), peak stress ( $\sim 23:30$  UTC), and the crisis bottom ( $\sim 02:00$  UTC March 11). At onset, BTC/USDC on Kraken shows modest bid-side dominance. By peak stress the imbalance is severe, with the bid curve towering over the ask across all 20 levels; this quantifies the flight from USDC at its most acute. By 02:00 UTC March 11, as the de-peg stabilized, the book had begun rebalancing.

BTC/USDT on Binance tells the opposite story. The bid and ask curves remain closely matched throughout all three moments, consistent with USDT absorbing inflows from participants completing the USDC $\rightarrow$ BTC $\rightarrow$ USDT rotation without generating comparable directional stress. The cross-pair contrast confirms that Bitcoin’s intermediary role was asymmetric: it absorbed distressed USDC on the bid side while passing a balanced flow into USDT.

### D. Synthesis and Regulatory Implications

These three findings combine into a coherent microstructure narrative. Spread widening confirmed that stress was deep and book-wide, not a top-of-book artifact; the bid-side depth surge quantified the scale of USDC exit pressure in real time; and the snapshot comparison documented how this pressure resolved through Bitcoin as an intermediary vehicle, leaving BTC/USDT

balanced while BTC/USDC was severely imbalanced at peak stress.

The perceived “flight-to-safety” market mechanism (USDC $\rightarrow$ BTC $\rightarrow$ USDT) has direct implications for the GENIUS Act’s reserve transparency requirements. The 12-hour-plus basis half-life documented in Section III reflects, in part, the microstructure friction visible here: when adverse selection risk is acute and order flow is one-directional, market makers widen quotes and the arbitrage mechanism that enforces parity breaks down. The reserve attestation requirements, outlined in the GENIUS Act, would reduce uncertainty about stablecoin backing; this would ideally dampen the severity of such one-sided flow episodes by compressing both the duration and magnitude of basis dislocations.

## VI. CONCLUSION

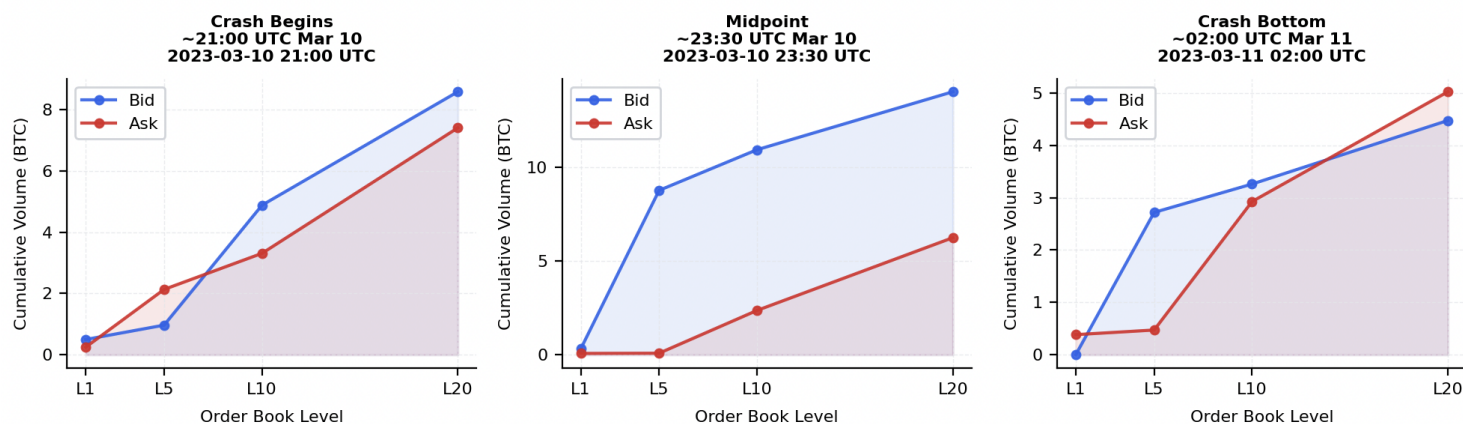
This paper has investigated the cross-currency dynamics of Bitcoin spot markets during the March 2023 USDC de-peg event, providing granular evidence on how stablecoin confidence breakdowns propagate through market microstructure. Our analysis yields three principal findings with direct implications for the emerging regulatory framework under the GENIUS Act.

### A. Conclusions and Implications for Trading, Risk Management, and Policy

*Implications for trading.* Cross-currency basis trades that appear risk-free under normal conditions can be-



### BTC/USDC Kraken — Order Book Depth Profile at Key Moments



### BTC-USDT Binance — Full 20-Level Order Book Depth

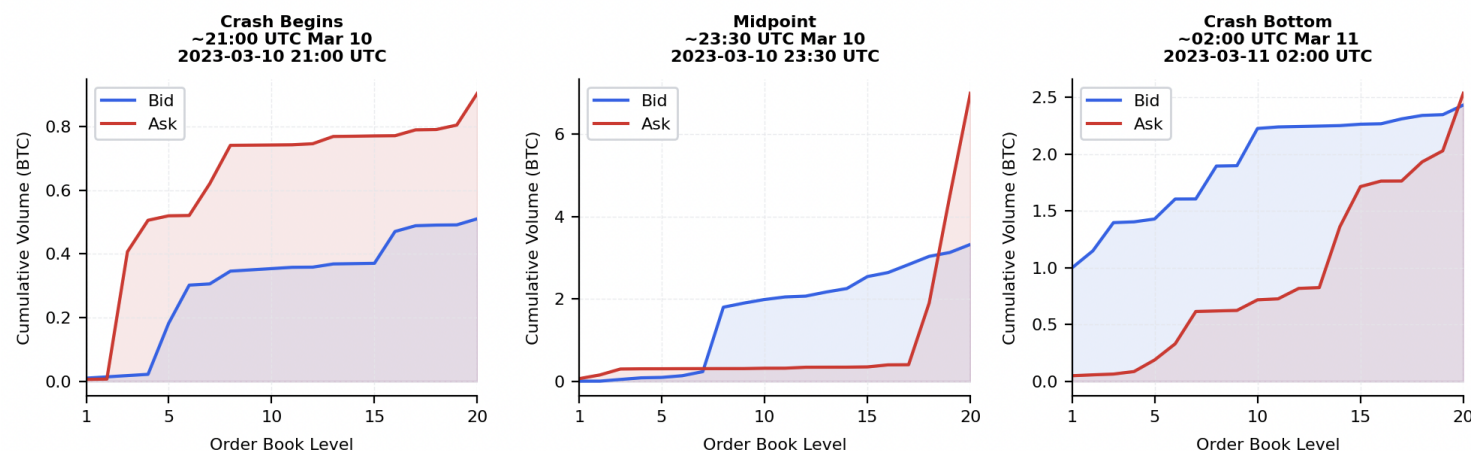


Fig. 5. Cumulative order book depth profiles (bid: blue, ask: red) at three crisis moments: onset (~21:00 UTC March 10), peak stress (~23:30 UTC), and crisis bottom (~02:00 UTC March 11). Top row shows BTC/USDC on Kraken; bottom row shows BTC/USDT on Binance. BTC/USDC exhibits severe bid-side dominance at peak stress, normalizing by the crisis bottom. BTC/USDT maintains approximate balance throughout.

come capital-intensive and slow to converge during reserve shocks. The expansion of half-lives from minutes to hours implies that arbitrageurs must hold inventory risk for materially longer horizons. Transaction-cost thresholds alone do not define profitability; persistence risk and order book imbalance become central determinants of execution viability.

*Implications for risk management.* Stablecoins should be treated as credit-sensitive instruments rather than mechanical dollar substitutes. The observed 14% peak dislocation and prolonged half-life demonstrate that basis risk can exceed typical crypto volatility assumptions. Institutional desks relying on stablecoin settlement, particularly for treasury operations or derivatives margins, should incorporate cross-currency basis risk into liquidity stress scenarios and consider diversification across issuers or direct fiat access.

*Implications for policy.* The GENIUS Act's emphasis on reserve transparency and banking-style supervision

directly addresses the uncertainty channel identified in this paper. The prolonged USDC basis deviations reflect uncertainty about reserve composition and recoverability. Standardized disclosure and oversight may compress the duration of such dislocations. At the same time, our evidence shows that stablecoin distress propagates instantly across cross-currency networks. Stablecoins function as funding currencies: when confidence weakens, fragmentation emerges globally within minutes. Regulatory frameworks may reduce issuer-specific risk, but they cannot eliminate the structural interconnectedness of crypto market microstructure.

### B. Regulatory Implications

These findings speak directly to the policy objectives motivating the GENIUS Act and broader stablecoin regulation. The legislation's emphasis on strict reserve backing, transparency, and banking-style oversight addresses precisely the vulnerability exposed by the SVB episode: when market participants cannot verify reserve

quality in real time, confidence can evaporate within hours and result in pricing dislocations that persist far longer than transaction costs alone could predict.

Our evidence suggests several specific implications for the regulatory framework:

*Reserve transparency requirements.* The observed half-life of 12+ hours for USDC basis deviations during stress reflects fundamental uncertainty about reserve composition and recover-ability. The GENIUS Act’s mandate for regular reserve attestations and clear disclosure of banking relationships would reduce this uncertainty. We speculate that had Circle’s SVB exposure been disclosed through standardized regulatory filings rather than an emergency Twitter announcement, market participants might have priced the risk more gradually. This would potentially avoid the extreme one-sided order flow that amplified the de-peg.

*Differential treatment of stablecoin issuers.* The persistent USDT premium documented in our recovery-period analysis (−25.8 bps from March 17–20) reflects a temporary reassessment of relative reserve quality that settled within the no-trade region defined by transaction costs. This “soft” premium suggests that markets discriminate among stablecoins based on perceived reserve safety even after acute stress subsides. Regulatory frameworks that impose uniform reserve standards across issuers may reduce such differentiation over time, potentially improving fungibility across stablecoin-quoted markets and tightening cross-currency bases.

*Systemic risk and contagion channels.* The DCC-GARCH evidence of crisis-induced correlation spikes demonstrates that stablecoin stress propagates rapidly across related markets. When USDC’s peg broke, BTC cross-pair pricing immediately reflected the same reserve-quality shock visible in direct USDC/USD markets, having no detectable lead-lag structure. This simultaneity implies that stablecoin distress cannot be contained within a single trading pair or venue; rather, it transmits instantaneously through the cross-currency network. Regulators designing systemic risk frameworks should recognize that stablecoins function as funding currencies whose distress affect all assets quoted against them.

*Market structure and fragmentation.* Our cross-exchange analysis reveals that the USDT safe-haven premium was remarkably consistent across Kraken, Binance, and Coinbase (within 1.4 bps during stress), indicating that despite the fragmented structure of cryptocurrency markets, information incorporation is highly efficient. This efficiency, however, operates in both directions: when confidence fails, the resulting basis dislocation propagates globally within minutes. The

GENIUS Act’s focus on regulated stablecoin issuers operating within a clear federal framework may facilitate faster arbitrage by reducing issuer-specific uncertainty, but it cannot eliminate the venue fragmentation that characterizes cryptocurrency trading.

*Implications for institutional adoption.* The Visa USDC settlement offering (and similar institutional initiatives) assume that regulated stablecoins will trade reliably at par. Our findings provide a cautionary counterpoint: even with federal oversight, extreme stress events can produce basis deviations exceeding 14% and order book imbalances that persist for hours. Institutional treasury desks and payment networks should incorporate stablecoin basis risk into their liquidity management frameworks, potentially holding reserves in multiple stablecoins or maintaining direct fiat access for stress scenarios.

### C. Limitations and Future Research

Our analysis focuses on a single stress event, and generalizability to other reserve-quality shocks remains an empirical question. Additionally, Coinbase data gaps during peak stress limited three-way venue comparison, and our one-minute order book frequency may miss sub-minute microstructure dynamics. Future work could extend this framework to the Terra/Luna collapse and FTX bankruptcy, examine whether basis dynamics tighten under the GENIUS Act regime, and link on-chain redemption mechanisms with centralized exchange order book dynamics.

### D. Concluding Remarks

The March 2023 USDC de-peg offers a natural experiment revealing that cryptocurrency markets, while highly efficient under normal conditions, are vulnerable to rapid and severe dislocations when stablecoin reserve quality is questioned. For market participants, stablecoins should be treated as credit instruments whose peg stability depends on reserve quality, not as riskless substitutes for fiat currency. For regulators, our evidence supports the GENIUS Act’s emphasis on reserve backing and disclosure as well as highlights the systemic nature of stablecoin distress. More broadly, the infrastructure connecting traditional banking to cryptocurrency trading, as exemplified by Circle’s SVB deposits, constitutes a critical transmission channel whose resilience merits ongoing attention.

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